

Contents

1	Brownian Motion in Soft Confinement	3
1.1	Introduction	3
1.2	Toy Model for soft Brownian Motion	4
1.2.1	Overview of the Toy Model	4
1.3	Theoretical developments	5
1.4	Numerical modelling	8
1.4.1	Discretisation	8
1.4.2	Time step selection	9
1.5	Results	10
1.5.1	Trajectories	10
1.5.2	Mean-Squared displacements	13
1.5.3	Quantification of diffusion enhancement	14
1.6	Conclusion and perspectives on the chapter	18

Chapter 1

Brownian Motion in Soft Confinement

1.1 Introduction

In many biological systems, one must not only take into account Brownian behaviours, but also carefully evaluate the effect of the system's or parts of the system's deformability on its dynamics. Indeed, if we take the example of a lipid bilayer, which is a soft and quasi-two-dimensional fluid, the thermal energy $k_B\Theta$ of the system is sufficient to cause long-wavelength bending fluctuations of the cell membrane, thickness fluctuations, lipid movement within the plane itself, and so forth. Naturally, a neighbouring fluid's motion is then affected by these shape modifications, which can in turn affect the dynamics of a body evolving in that same bath. This coupling is not trivial.

The coupling between softness and viscous flows, which we will refer to as the elasto-hydrodynamic (EHD) coupling, has been extensively studied in the deterministic framework. We for example mention the emergence of lift forces near a deformable boundary at low Reynolds numbers, which have been first demonstrated theoretically [12] [1][14] [15] [16] and then experimentally [11] [2] [10] [17] [18].

The field of stochastic EHD is still fairly new. Molecular dynamics simulations have been conducted to investigate the anomalous diffusion of a micelle near a lipid bilayer [13]. Another recent study [3] demonstrated experimentally the emergence of a new, non-conservative force acting upon a Brownian soft colloid near a confining wall. However, we still to this day lack a precise theory to encompass quantitatively the long-time signatures of the coupling between softness, confinement and fluctuations.

Because of the level of complexity of such a study, I present in this chapter a most simple model, with elementary ingredients, which can still qualitatively capture all the effects we are interested in. This model was published in *Communications Physics* in 2025 [6] and the full print and Supplementary Information can be found in appendix of this thesis. In the following sections, I present the theoretical aspects of the model, developed by David S. Dean. I develop the numerical model used to simulate this toy model. Finally, I show how numerics and theory compare.

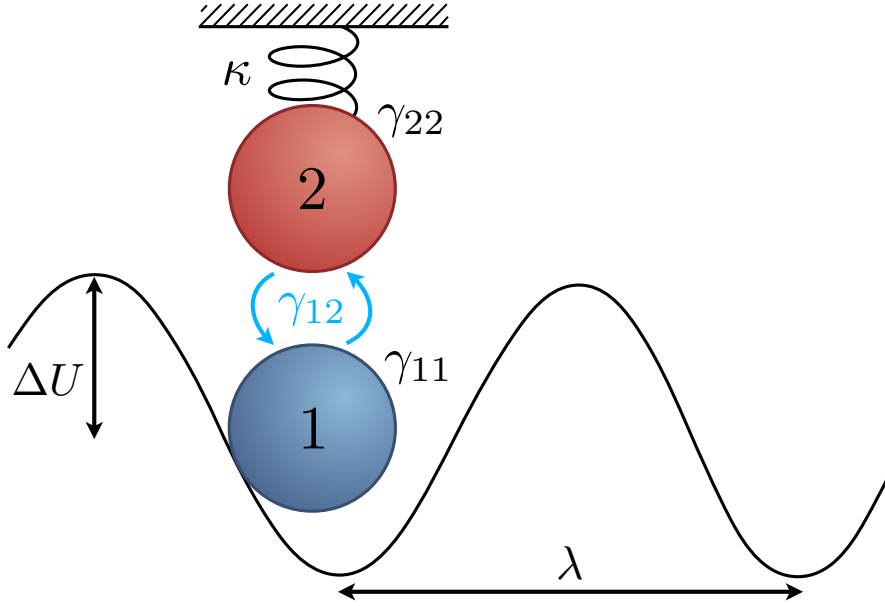


Figure 1.1: Representation of the system. A Brownian colloid of interest (blue) is trapped in a periodic potential of amplitude ΔU and spatial period λ . An elastic membrane is modelled locally as a second Brownian colloid (red) subject to a harmonic potential of inverse stiffness (or *compliance*) κ . This membrane is coupled to the colloid of interest through an indirect, hydrodynamic-like coupling represented by the coefficient γ_{12} , and both the colloid and the membrane feel their own self-friction γ_{11} and γ_{22} , respectively.

1.2 Toy Model for soft Brownian Motion

1.2.1 Overview of the Toy Model

From here on, we study a purely one-dimensional case of a Brownian colloid coupled to an elastic and fluctuating membrane (Fig. 1.1). The Brownian colloid is represented by a first degree of freedom, q_1 , and is submitted to a periodic trapping potential $\phi(q_1)$ of spatial period λ and amplitude ΔU . Throughout this chapter, the Brownian colloid's degree of freedom q_1 will be color-coded in blue, and the membrane's degree of freedom q_2 in red. We couple to this colloid the toy model version of the membrane. We represent the membrane with a colloid that is submitted in a harmonic potential of energy $\frac{q_2^2}{2\kappa}$, where κ is the compliance (or inverse stiffness). It represents the elasticity of the membrane. This second colloid is tracked with the second degree of freedom, q_2 .

Both the Brownian colloid and the membrane feel their own self-friction, respectively γ_{11} and γ_{22} . Furthermore, they interact through an indirect, hydrodynamic-like interaction γ_{12} . This interaction represents the effect of the fluid that is displaced by the motion of the colloid on the membrane, and *vice-versa*. All together, these terms form

the friction matrix $\mathbf{\Gamma}$

$$\mathbf{\Gamma} = \begin{pmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{12} & \gamma_{22} \end{pmatrix} \quad (1.1)$$

which can be inverted to give the mobility matrix of the system, as $\mathbf{M} = \mathbf{\Gamma}^{-1}$. As per Lorentz's reciprocal theorem, both matrices must be positive definite, and, because their diagonal terms account for the self-friction and self-mobility respectively, $\mu_{11} > 0$ and $\mu_{22} > 0$.

We can write the total energy of the system as

$$H(q_1, q_2) = \phi(q_1) + \frac{q_2^2}{2\kappa}, \quad (1.2)$$

and we notice the absence of Hamiltonian interactions in the system. In that case, we work in the overdamped limit, and we write the coupled Langevin equations describing the evolutions of q_1 and q_2 as

$$\gamma_{11} \frac{dq_1}{dt} + \gamma_{12} \frac{dq_2}{dt} = -\phi'(q_1) + \eta_1(t), \quad (1.3)$$

$$\gamma_{12} \frac{dq_1}{dt} + \gamma_{22} \frac{dq_2}{dt} = -\frac{q_2}{\kappa} + \eta_2(t), \quad (1.4)$$

where η_i and η_j (with $j \in \{1, 2\}$) are Gaussian white noises of zero mean and correlation functions $\langle \eta_i(t) \eta_j(t') \rangle = 2k_B T \delta(t - t') \gamma_{ij}$. The equilibrium probability density function of the system is then given by the Gibbs-Boltzmann distribution, as

$$P_{\text{eq}} = \frac{\exp[-\beta H(q_1, q_2)]}{Z}. \quad (1.5)$$

1.3 Theoretical developments

The objective of the following developments is to compute the modification to the long-time dynamics of the Brownian colloid diffusing within the sinusoidal trap. The development I choose to present here is not the most rigorous, and corresponds to the very stiff ($\kappa \rightarrow 0$) limit of the elastic membrane. The interested reader can find a more rigorous argument based on multiscale analysis [5] [9] in the appendices of this thesis, alongside a second development for the very compliant limit ($\kappa \rightarrow \infty$).

We begin by refactoring Eqs. 1.3 and 1.4 in the deterministic case $\Theta = 0$

$$\frac{dq_1}{dt} = -\mu_{11} \phi'(q_1) - \frac{\mu_{12}}{\kappa} q_2, \quad (1.6)$$

$$\frac{dq_2}{dt} = -\mu_{12} \phi'(q_1) - \frac{\mu_{22}}{\kappa} q_2. \quad (1.7)$$

We then integrate to express q_2 as

$$q_2 = -\frac{\kappa \mu_{12}}{\mu_{22}} \left(1 + \frac{\kappa}{\mu_{22}} \frac{d}{dt} \right)^{-1} \phi'(q_1). \quad (1.8)$$

In the small κ limit, we then expand the right-hand bracket term, taking into account the chain rule

$$q_2 \simeq -\frac{\kappa\mu_{12}}{\mu_{22}} \left[\phi'(q_1) - \frac{\kappa}{\mu_{22}} \phi''(q_1) \frac{dq_1}{dt} + O(\tau_2^2) \right] . \quad (1.9)$$

Here, we note that τ_2 is the characteristic relaxation time of the harmonic trapping corresponding to the membrane elasticity, as $\tau_2 = \frac{\kappa}{\mu_{22}}$. Now, substituting this new expression for q_2 in Eq. 1.6, we obtain

$$\frac{dq_1}{dt} = - \left(\mu_{11} - \frac{\mu_{12}^2}{\mu_{22}} \right) \phi'(q_1(t)) - \frac{\kappa\mu_{12}^2}{\mu_{22}^2} \phi''(q_1(t)) \frac{dq_1}{dt} , \quad (1.10)$$

which we can reexpress as

$$\frac{dq_1}{dt} = -\mu_e(q_1) \phi'(q_1) , \quad (1.11)$$

where μ_e is a dressed mobility with expression

$$\mu_e = \frac{\mu_{11}\mu_{22} - \mu_{12}^2}{\mu_{22} \left(1 + \frac{\kappa\phi''(q_1)\mu_{12}^2}{\mu_{22}^2} \right)} . \quad (1.12)$$

Remembering that $\mathbf{\Gamma} = \mathbf{M}^{-1}$,

$$\mu_e = \frac{1}{\gamma_{11} \left(1 + \frac{\kappa\phi''(q_1)\mu_{12}^2}{\mu_{11}^2} \right)} . \quad (1.13)$$

Now, we can avoid reexpressing the noise which should be applied to Eq. 1.11 for the thermal case by working directly with Fokker-Planck formalism. This formalism is introduced in its elementary form at the beginning of my thesis, and developed for more complex cases in Chapter 2. **CHECK CHAPT NR AT THE END OF THE REDACTION HERE** Doing such a thing is not rigorous, as I previously mentioned, because the effective stochastic process has not been derived. Eliminating fast variables in Eq. 1.11 typically introduces non-trivial corrections. However, since we assume separation of timescales, the effective slow process still permits the writing of a uniquely-defined Fokker-Planck equation.

This Fokker-Planck equation reads

$$\frac{\partial P_1(q_1, t)}{\partial t} = \frac{\partial}{\partial q_1} \mu_e(q_1) \left[k_B \Theta \frac{\partial P_1(q_1, t)}{\partial q_1} + \phi'(q_1) P_1(q_1, t) \right] , \quad (1.14)$$

and, noting that the time derivative of the probability density function $P_1(q_1, t)$ can be expressed in terms of probability flux J ,

$$\frac{\partial P_1(q_1, t)}{\partial t} = -\frac{\partial J}{\partial q_1} . \quad (1.15)$$

This allows us to write

$$J = -D_e(q_1) \frac{\partial}{\partial q_1} P_1(q_1, t) - \mu_e \phi'(q_1) P_1(q_1, t) , \quad (1.16)$$

where $D_e = k_B \Theta$. The above equation can be rewritten

$$J = -D_e(q_1) \exp[-\beta\phi(q_1)] \frac{d}{dq} \{ \exp[\beta\phi(q_1)] P_1(q_1, t) \} \quad (1.17)$$

with $\beta = \frac{1}{k_B \Theta}$.

Now, we tilt the potential by adding a small force F , which gives a new potential $U(q_1)$, as

$$U(q_1) = \phi(q_1) - Fq_1 . \quad (1.18)$$

Then, the current equation can be rewritten taking into account this new force, and gives

$$\frac{d}{dq_1} \{ \exp[\beta U(q_1)] P(q_1, t) \} = -\frac{J(q_1, t)}{D_e(q_1)} \exp[\beta U(q_1)] . \quad (1.19)$$

We can then integrate over a period λ , and, using periodicity, we retrieve

$$P_1(q_1, t) = \frac{J(q_1) \exp[-\beta U(q_1)]}{1 - \exp(-\beta F \lambda)} \int_{q_1}^{q_1 + \lambda} \frac{\exp[\beta U(q)]}{D_e(q)} dq . \quad (1.20)$$

By essence, $P_1(q_1, t)$ is normalised

$$\int_0^\lambda P_1(q_1) dq_1 = 1 , \quad (1.21)$$

and thus we retrieve for J

$$J = \frac{1 - \exp[-\beta F \lambda]}{\int_0^\lambda \exp[-\beta U(q)] dq \int_q^{q+\lambda} \frac{\exp[\beta U(q)]}{D_e(q)} dq} . \quad (1.22)$$

In the small force limit $F \rightarrow 0$,

$$1 - \exp(-\beta F \lambda) \simeq \beta F \lambda , \quad (1.23)$$

which, to leading order in F , gives

$$J = \frac{\beta F \lambda}{\int_0^\lambda \exp[-\beta\phi(q_1)] dq_1 \int_0^\lambda \frac{\exp[\beta\phi(q_1)]}{D_e(q_1)} dq_1} . \quad (1.24)$$

Finally, we note that the drift velocity should read $v = J(q_1)\lambda$, and that the effective diffusion coefficient over λ should be

$$D^* = \frac{k_B \Theta v}{F} = \frac{\lambda^2}{\int_0^\lambda \exp[-\beta\phi(q_1)] dq_1 \int_0^\lambda \frac{\exp(\beta\phi(q_1))}{D_e(q_1)} dq_1} , \quad (1.25)$$

which is the generalised Lifson-Jackson formula [7].

Finally, we replace $D_e(q_1)$ in the expression

$$D^* = \frac{k_B \Theta \lambda^2}{\gamma_{11} \int_0^\lambda dq \exp[\beta \phi(q)] \left(1 - \frac{\kappa \gamma_{12}^2 \phi'^2(q)}{k_B \Theta \gamma_{11}^2}\right) \int_0^\lambda dq \exp[-\beta \phi(q)]} . \quad (1.26)$$

And, setting $\phi(q_1) = \Delta U \phi(\frac{q_1}{\lambda})$, where ΔU is the amplitude of the periodic trap

$$D^* = \frac{k_B \Theta}{\gamma_{11} \int_0^1 dp \exp[\beta \Delta U \psi(p)] \left(1 - \frac{\kappa \gamma_{12}^2 \Delta U^2 \psi'^2(p)}{k_B \Theta \gamma_{11}^2}\right) \int_0^1 dp \exp[-\beta \Delta U \psi(p)]} . \quad (1.27)$$

Which is the equation we use to compare with numerical results.

In the $\kappa \rightarrow 0$ limit, the above expression can be approximated as

$$D^* = D^*(0)(1 + \alpha) , \quad (1.28)$$

with

$$\alpha = \beta \epsilon \frac{\lambda^2 \gamma_{12}^2}{\gamma_{11}^2} \frac{\int_0^\lambda dq_1 \phi'(q_1)^2 \exp[\beta \phi(q_1)]}{\int_0^\lambda dq_1 \exp[-\beta \phi(q_1)]} , \quad (1.29)$$

where we define $\epsilon = \frac{k_B \Theta}{\lambda^2}$ a dimensionless compliance. As $\alpha > 0$, the coupling of the membrane to the Brownian colloid always increases its long-time diffusion constant.

Furthermore, in the very compliant limit $\kappa \rightarrow \infty$, D^* becomes

$$D^* = k_B \Theta \frac{\mu_{11}(\mu_{11}\mu_{22} - \mu_{12}^2)}{\lambda^{-2} Z_+ Z_- (\mu_{11}\mu_{22} - \mu_{12}^2) + \mu_{12}^2} + O\left(\frac{1}{\sqrt{\epsilon}}\right) . \quad (1.30)$$

We notice, in that regime, the absence of a κ or ϵ dependence.

1.4 Numerical modelling

1.4.1 Discretisation

My main mission in this project is to simulate the above Eqs. 1.3 and 1.4 efficiently. To do so, I discretise these equations using a simple forward Euler scheme (which was introduced in the beginning of this thesis). The equations read

$$q_1(t + \Delta t) - q_1(t) = -\mu_{11} \phi'(q_1(t)) \Delta t - \frac{\mu_{12}}{\kappa} q_2(t) \Delta t + \xi_1 \sqrt{2k_B T \Delta t} , \quad (1.31)$$

$$q_2(t + \Delta t) - q_2(t) = -\frac{\mu_{22}}{\kappa} q_2(t) \Delta t - \mu_{12} \phi'(q_1(t)) \Delta t + \xi_2 \sqrt{2k_B T \Delta t} , \quad (1.32)$$

where ξ_i is a zero-mean Gaussian random variable, as

$$\langle \xi_i \xi_j \rangle = \mu_{ij} . \quad (1.33)$$

We choose to work in terms of mobility instead of friction matrix elements for convenience in the upcoming developments. They are, however, strictly equivalent.

Furthermore, in the rest of this study, we choose $\phi(q_1)$ to be sinusoidal, as $\phi(q_1) = \Delta U \sin\left(\frac{2\pi q_1}{\lambda}\right)$, with ΔU the amplitude of the sinusoidal trap.

1.4.2 Time step selection

In any discretisation process, it is essential to choose the appropriate time step. This time step must be small enough to avoid a blowup from error propagation (again, see the beginning of this thesis for more details on how the Euler method propagates errors), but big enough to remain advantageous. An arbitrarily small timestep prohibits the probing of very long time dynamics due to computation time.

To select a correct time step for a simulation, I typically first identify all the smaller time scales in the system I want to simulate. In the present study, I identify two time scales : the local relaxation time within a local minimum of the sinusoidal trap, and the relaxation time of the harmonic trap.

Near a local minimum, we approximate the sinusoidal trap as harmonic. In that case,

$$\phi(q_1) = \frac{1}{2}k_{\text{eff}}q_1^2, \quad (1.34)$$

where k_{eff} is an effective stiffness, which we identify as the local curvature of the potential as

$$k_{\text{eff}} = |\phi''(0)| \quad (1.35)$$

$$= \Delta U \left(\frac{2\pi}{\lambda} \right)^2, \quad (1.36)$$

and we can finally identify this first timescale as

$$\tau_1 = (\mu_{11}k_{\text{eff}})^{-1} = \frac{\lambda^2}{4\pi^2\Delta U\mu_{11}}. \quad (1.37)$$

The harmonic trap is much simpler, as it requires no approximations, and we find its characteristic timescale

$$\tau_2 = \frac{\kappa}{\mu_{22}}. \quad (1.38)$$

Then, I choose a time step which satisfies

$$\Delta t = \frac{1}{100} \max \tau_1, \tau_2. \quad (1.39)$$

We also need to define an upper bound to the simulation time. We must always remain mindful of the computational cost of our simulations, as time and energy both are scarce resources. Thus, we can define a timescale τ^* which depends on the long-time effective diffusion coefficient $D^*(\kappa)$ defined in Eq. 1.27

$$\tau^*(\kappa) = \frac{\lambda^2}{D^*(\kappa)}. \quad (1.40)$$

I choose to run my simulations for a time T_{sim} which is much larger than τ^* , as $T_{\text{sim}} = 100\tau^*$. This allows me to probe the long-time dynamics of the system, and to extract the effective diffusion coefficient from the mean-squared displacement of the Brownian

colloid. This, in turn, sets the computational cost of a simulation, which I can quantify using the total number of iterations in the time loop N as

$$N = \left\lfloor \frac{T_{\text{sim}}}{\Delta t} \right\rfloor = \left\lfloor \frac{100\tau^*}{\Delta t} \right\rfloor. \quad (1.41)$$

1.5 Results

The interested reader can find a minimal version of a trajectory generation code [here](#). This simple code allows the user to tune parameters and test for themselves their effect on the dynamics of the system. Most of the figures presented in this section are generated using this simpler script, in which I left the seed fixed to ensure proper reproducibility.

1.5.1 Trajectories

I first generate trajectories of the two coupled Brownian colloids using Eqs. 1.31 and 1.32, for a given set of parameters. A sample of a single trajectory is shown in Fig. 1.2. We can see that the membrane fluctuates around its equilibrium position in $q_2 = 0$, whereas the Brownian colloid is trapped at short times within a local minimum, and escapes through random thermal fluctuations at longer times to explore space. This is a first verification that the simulations behave as expected. Indeed, it is not unusual for Brownian simulations to blowup in the case of an unfortuitous dimensionalisation of the system, and this issue is almost always imputable to a too-large time step. We have already discussed this matter in the beginning of the manuscript.

We verify the behaviour of the system at equilibrium by sampling the probability density function of q_1 and q_2 from the generated trajectories, and comparing them to the theoretical prediction from the Gibbs-Boltzmann distribution (Eq. 1.5). This is shown in Fig. 1.3. We can see that the sampled probability density functions (histograms) and theoretical prediction from the Gibbs-Boltzmann distribution (solid lines, Eq. 1.5) are in perfect agreement, even for low statistics (a single binned trajectory of 10^8 points suffices to reach a good match between theory and numerics).

The sampled probability distribution of q_1 follows the expected periodic shape, whereas the one of q_2 follows the expected Gaussian shape predicted by Eq. 1.5. This is a second verification that the simulations are working as expected, and that we are correctly sampling the equilibrium distribution of the system.

However, at this stage, we have not probed the dynamics of the system. Thus, we still have no insight on the effect of a soft membrane on the diffusion of the Brownian colloid. To do so, we will need to compute the MSD of q_1 and q_2 from the generated trajectories, and to extract the effective diffusion coefficient of q_1 from the long-time behaviour of its MSD. We will then be able to compare this effective diffusion coefficient to the theoretical prediction from Eq. 1.27.

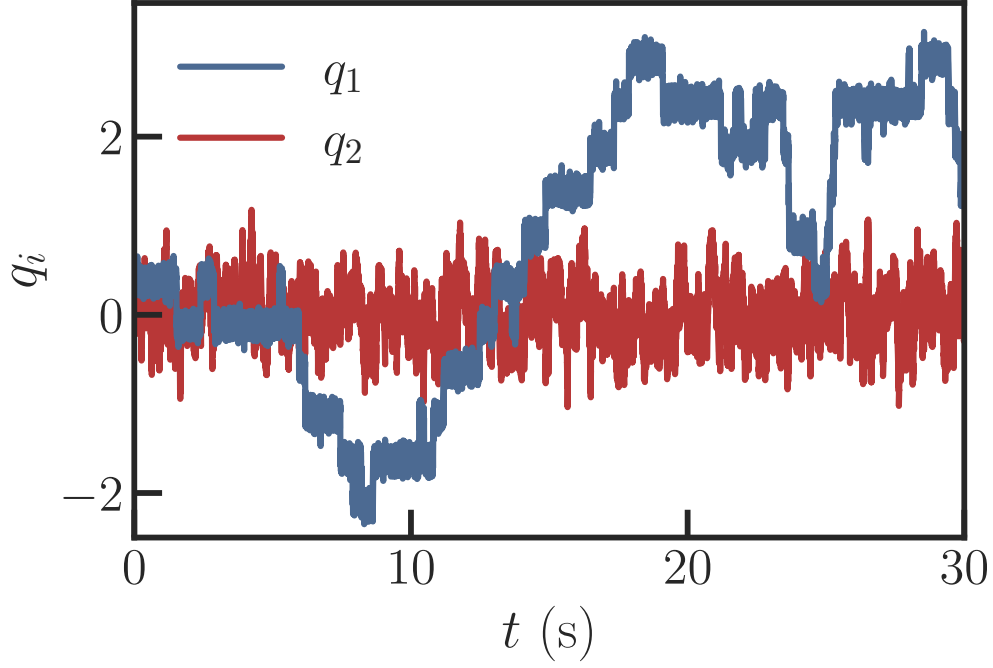


Figure 1.2: Trajectories of the Brownian colloid (blue) and the membrane (red) for $\kappa = 0.01$. The other parameters are fixed to $\Delta U = 2.7k_B\Theta$, $k_B\Theta = 1.0$, $\lambda = 1.0$, $\gamma_{11} = 1.0$, $\gamma_{22} = 1.0$ and $\gamma_{12} = 0.8$, $\tau = 10^{-4}$. The membrane noticeably fluctuates around its equilibrium position in $q_2 = 0$, whereas the colloid is trapped at short times within a local minimum, and escapes through random thermal fluctuations at longer times to explore available space.

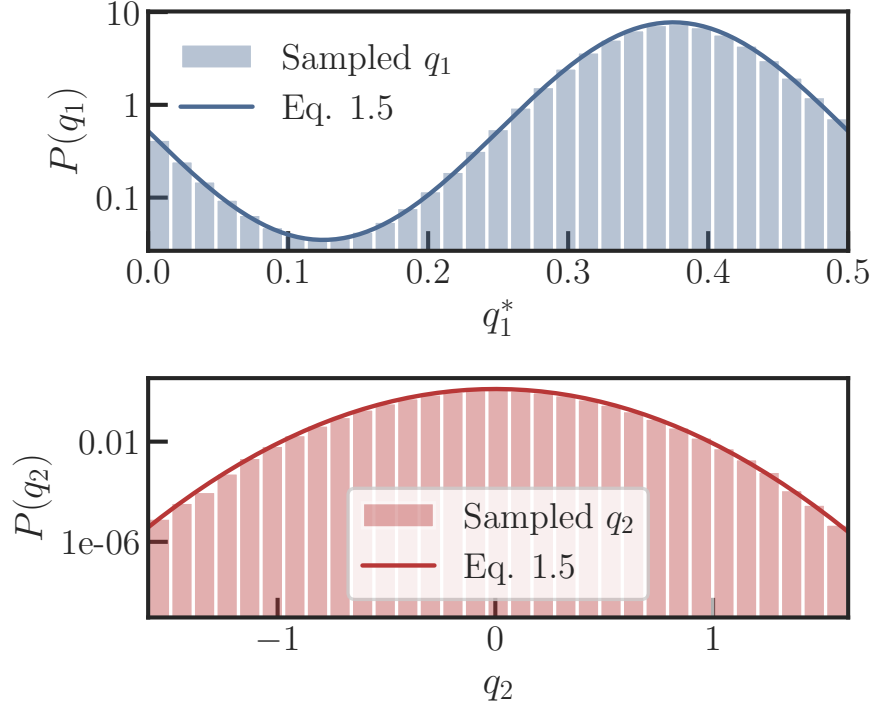


Figure 1.3: Equilibrium probability density function of q_1 (top, for $q_1^* = q_1 \bmod \lambda$) and q_2 (bottom) for $\kappa = 0.01$. The other parameters are fixed to $\Delta U = 2.7k_B\Theta$, $k_B\Theta = 1.0$, $\lambda = 1.0$, $\gamma_{11} = 1$, $\gamma_{22} = 1$, $\gamma_{12} = 0.8$, $\tau = 10^{-4}\text{s}$, for a single trajectory of $N = 1 \times 10^7$ points. The sampled probability density functions (histograms) and theoretical prediction from the Gibbs-Boltzmann distribution (Eq. 1.5, solid lines) are in perfect agreement, even for a single trajectory of 10^8 points.

1.5.2 Mean-Squared displacements

In the present case below, the MSD is computed directly from the generated trajectories, calculating the average of the squared displacements over all possible or given time lags. This means that the computer will store the whole trajectories q_1 and q_2 in memory. It will then loop over all possible time lags Δt , and for each of them, it will compute the average of the squared displacements $[q(t + \Delta t) - q(t)]^2$ over all possible time origins t . This technique is straightforward and sufficient for demonstration purposes.

Listing 1.1: Direct computation of the MSD from trajectories

```
def MSD(x, lags):
    msd = np.zeros(len(lags)) # Always initialise you arrays if
                              # you know their size beforehand.

    for n, lag in enumerate(lags):
        if lag >= len(x):      # Avoid pitfalls.
            msd[n] = np.nan

        else:
            msd[n] = np.nanmean( # Ignore NaN values !
                                (x[:-lag] - x[lag:])**2 # In words, NaN-dropped
                                mean of the square of displacements for timelag
                                lag of index n in the lag array.
                                )

    return msd # Returns an array of size n with the MSD for each
              # time lag in the lags array.
```

With a simple estimate looking at the code above, we can see that the computational cost of such a method scales as $O(N^2)$ at worst.

In our published work and in similar numerical studies [8], it is more convenient to first compute the *histograms of displacements* at given time lags. We can then simply extract the MSD from the width of the histogram of displacements, as

$$\text{MSD}(\Delta t) = \sigma^2(\Delta t) , \quad (1.42)$$

where σ^2 is the variance of the histogram of displacements at time lag Δt .

Though not computationally lighter, this is the technique I recommend for this type of statistical study for the reasons that follow. The histogram of displacements has the advantage of being a stackable quantity across iterations of a same simulation. On a rolling basis, as multiple trajectories are calculated, we simply add counts to the histogram of displacements. Furthermore, it is a much more general quantity than the MSD, as it allows for the computation of other moments of the distribution of displacements (skewness, kurtosis, etc.).

We can define a convergence criterion on the histogram of displacements as the relative distance between a "previous" histogram of cumulative displacements, and an "updated" histogram which contains all previous iteration plus the new additions from

the latest simulated trajectories. When this relative distance is smaller than a given threshold for a sufficient number of iterations, we deem the simulation to have converged, and it automatically stops. This is a convenient way of saving resources and active working time, as it allows full automation of the whole process.

I define the convergence criterion as

$$\varepsilon = \frac{\sum_i |P_i^{\text{prev}} - P_i^{\text{updated}}|}{\sum_i P_i^{\text{prev}}} , \quad (1.43)$$

where P is the histogram of displacements, and i the index of the bin. I choose a threshold of $\varepsilon \leq 5 \times 10^{-3}$ for more than 3 iterations of 10 stacked trajectory statistics. This means that, when the relative distance between the "previous" and "updated" histograms of displacements is smaller than 5×10^{-3} for more than 3 iterations of 10 stacked trajectories, we consider the simulation to have converged, and it automatically stops.

The results of a typical MSD calculation for q_1 and q_2 are shown in Figs. 1.4 and 1.5. We can see that the MSD of q_1 (Fig. 1.4) is at first diffusive with a Δt exponent of 1 and a diffusion coefficient D_m which is directly proportional to the self-mobility of the Brownian colloid as

$$D_m = k_B \Theta \mu_{11} . \quad (1.44)$$

It then falls into a slow crossover to reach a second diffusive regime of effective diffusivity D^* as defined in Eqs. 1.27 and 1.30 for the very rigid and very compliant cases. The long-time diffusion coefficient is smaller than the short-time diffusion coefficient, which is a signature of the trapping effect of the sinusoidal potential.

The MSD of q_2 (Fig. 1.5) is also diffusive at short times with a diffusion coefficient D_2 which is directly proportional to the self-mobility of the membrane as

$$D_2 = k_B \Theta \mu_{22} , \quad (1.45)$$

and then crosses over to a confined regime at long times $\Delta t > \tau_2$ due to the harmonic trapping of the membrane. We can predict the value of the long-time MSD plateau of q_2 as $\text{MSD}(q_2) = 2k_B \Theta \kappa$, which we verify numerically.

1.5.3 Quantification of diffusion enhancement

We can finally infer all diffusion coefficients by performing affine fits of the MSDs in the diffusive regimes, remembering that in the diffusive regime

$$\text{MSD}(\Delta t) = 2D\Delta t . \quad (1.46)$$

In log scale, a diffusive regime corresponds to a linear regime with a time exponent of 1 as $\log(\text{MSD}) = \log(2D) + \log(\Delta t)$, and the diffusion coefficient can be directly inferred from the intercept of the affine fit to the MSD axis at the origin of Δt .

For a given set of parameters, we can then compare the effect of a varying compliance of a soft membrane on the long-time diffusion coefficient of a Brownian colloid trapped nearby in a sinusoidal potential (Fig. 1.6).

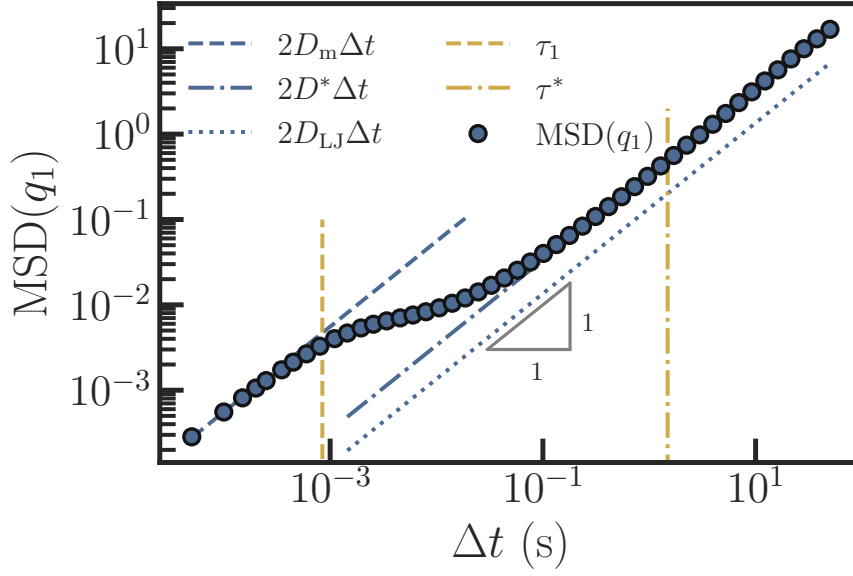


Figure 1.4: Mean-Squared Displacement of q_1 for $\kappa = 0.01$, $\Delta U = 2.7k_B\Theta$, $k_B\Theta = 1.0$, $\lambda = 0.5$, $\gamma_{11} = 1$, $\gamma_{22} = 1$, $\gamma_{12} = 0.8$, $\tau = 10^{-4}\text{s}$, for a single trajectory of $N = 1 \times 10^8$ points. At small timelags $\Delta t < \tau_1$, the Brownian colloid presents a diffusion constant D_m (Eq. 1.44, in blue dashes) which corresponds to a bulk, free diffusivity inside a local minimum of the potential $\phi(q_1)$. It then slowly crosses over to a second diffusive regime for $\Delta t > \tau^*$ with diffusion constant D^* (Eq. 1.30, in blue dashdot). The dotted blue line indicates the expected long-time diffusion coefficient in the very rigid limit $\kappa \rightarrow 0$ (Eq. 1.27 for $\kappa = 0$), which is smaller, as expected.

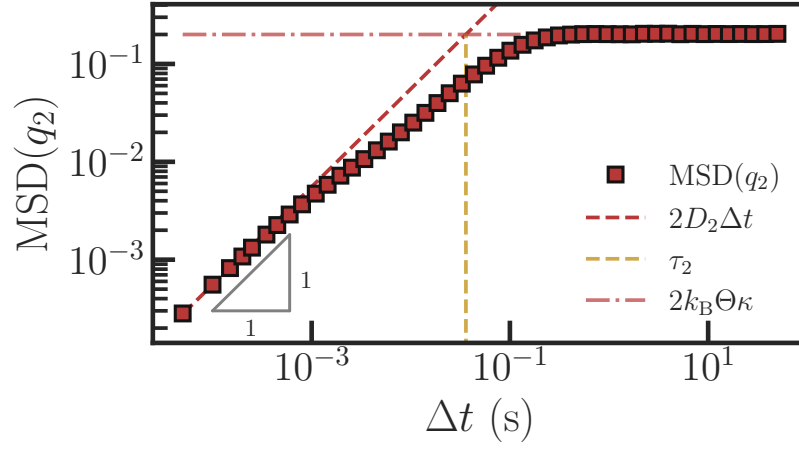


Figure 1.5: Mean-Squared Displacement of q_2 for $\kappa = 0.01$, $\Delta U = 2.7k_B\Theta$, $k_B\Theta = 1.0$, $\lambda = 0.5$, $\gamma_{11} = 1$, $\gamma_{22} = 1$, $\gamma_{12} = 0.8$, $\tau = 10^{-4}\text{s}$, for a single trajectory of $N = 1 \times 10^8$ points. At small timelags $\Delta t < \tau_2$, the membrane presents a diffusion constant D_2 (Eq. 1.45, in red dashes) which corresponds to a bulk, free diffusivity inside the minimum of the harmonic potential. It then slowly crosses over to a second diffusive regime for $\Delta t > \tau^2$ with constant MSD. The dotted red line indicates the expected plateau value for the long-time MSD at $2k_B\Theta\kappa$.

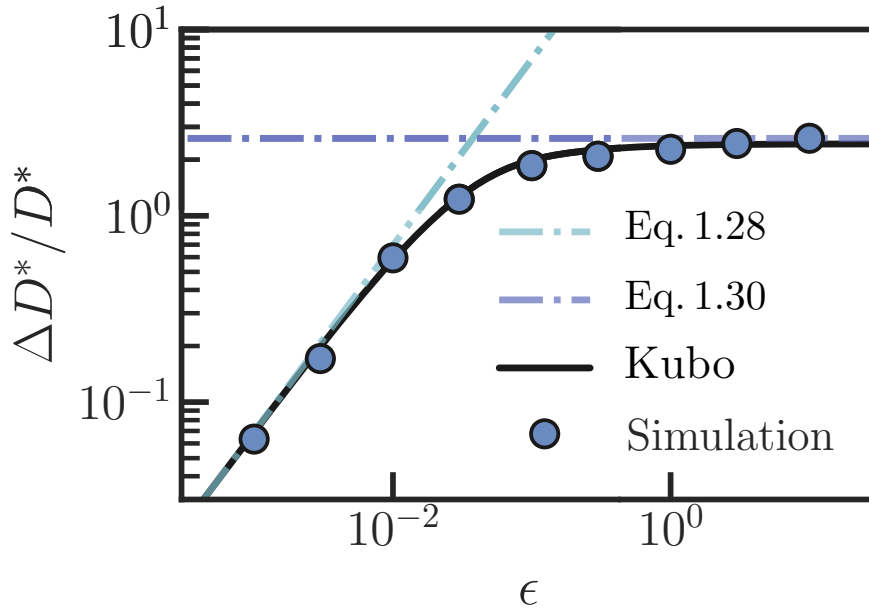


Figure 1.6: Augmentation factor $\frac{\Delta D^*(\epsilon)}{D^*}$ as a function of the coupling parameter ϵ for $\Delta U = 3.0k_B\Theta$, $k_B\Theta = 1.0$, $\lambda = 1.0$, $\mu_{11} = 1$, $\mu_{22} = 1$, $\mu_{12} = 0.85$. The numerical results (blue dots) are in perfect agreement with the expected theoretical augmentation factors in the asymptotical regions $\epsilon \rightarrow 0$ and $\epsilon \rightarrow \infty$ (dash-dotted lines) of Eqs. as well as the crossover region, which is accessed by solving the full two-dimensional Kubo problem (solid black line).

We define an augmentation factor $\frac{\Delta D^*(\epsilon)}{D^*}$ as

$$\frac{\Delta D^*(\epsilon)}{D^*} = \frac{D^*(\epsilon) - D^*(0)}{D^*(0)}, \quad (1.47)$$

to conveniently compare the coupled to uncoupled case. We notice that numerical results seem in perfect agreement with the expected theoretical augmentation factors in the asymptotical regions $\epsilon \rightarrow 0$ and $\epsilon \rightarrow \infty$. Interestingly, one may think of ϵ as a ratio of timescales. At very small ϵ , the membrane is extremely stiff, and q_2 becomes the fast mode. At very large ϵ , on the contrary, it comparatively becomes the slow mode.

Around $\epsilon = O(1)$, however, the timescale separation is lost. The particle coordinate and the elastic degree of freedom now relax on comparable timescales, and thus, neither Lifson-Jackson expression is valid.

To access this crossover regime, we solve the full two-dimensional Kubo problem¹[4]. All the relevant developments and details on the numerical solution of the Kubo equation can be found in the appendix of this thesis. I will not go into details here, as I did not personally work on that matter.

This numerical solution does not require assuming either $\epsilon \ll 1$ or $\epsilon \gg 1$, and therefore gives the crossover between the two asymptotic regimes. It is, however, much more tedious to estimate due to the computational cost of solving.

Furthermore, we may input to the model biological values for the parameters. Let us consider a colloid of radius $a = 100\text{nm}$ at a distance $d = 10\text{nm}$ from the membrane of elastic modulus $E = 1\text{kPa}$. The equivalent compliance of the membrane can be found using Hertz's model to give $\kappa = (E\sqrt{ad})^{-1}$. The system is immersed in a fluid of viscosity $\eta = 10^{-3}\text{Pa.s}$, at room temperature $T = 300\text{K}$. The Brownian colloid is subject to a periodic potential of amplitude $\Delta U = 4k_B\Theta$ and spatial period $\lambda = a = 100\text{nm}$.

To recover mobility values, we assume $\mu_{11} = mu_{22} = (6\pi\eta a)^{-1}$, and assume Oseen's mobility expression for μ_{12} as $\mu_{12} = (4\pi\eta r)^{-1}$, with r the center-to-center distance between the Brownian colloid and the colloid representing the membrane, as $r = 2a + d$.

With these values, the adimensionalised compliance ϵ falls exactly in the crossover region where the system is not considered very stiff or very compliant. Still, we may estimate the expected enhancement of the long-time diffusion coefficient by simply plotting these two asymptotes (Fig. 1.7) and recover at $\epsilon = 0.013$ an expected enhancement of about $\frac{\Delta D^*}{D^*} \approx 100\%$.

1.6 Conclusion and perspectives on the chapter

In this Chapter, we have exposed the novelty of the field of stochastic elastohydrodynamics. We have developed a toy model with elementary ingredients which encompasses qualitatively the physics of a Brownian colloid diffusing nearby a soft membrane. We have then developed a theoretical framework to predict the long-time diffusion coefficient of the Brownian colloid as a function of the compliance of the membrane, and

¹for which the interested reader can find the FlexPDE code in appendix of this manuscript.

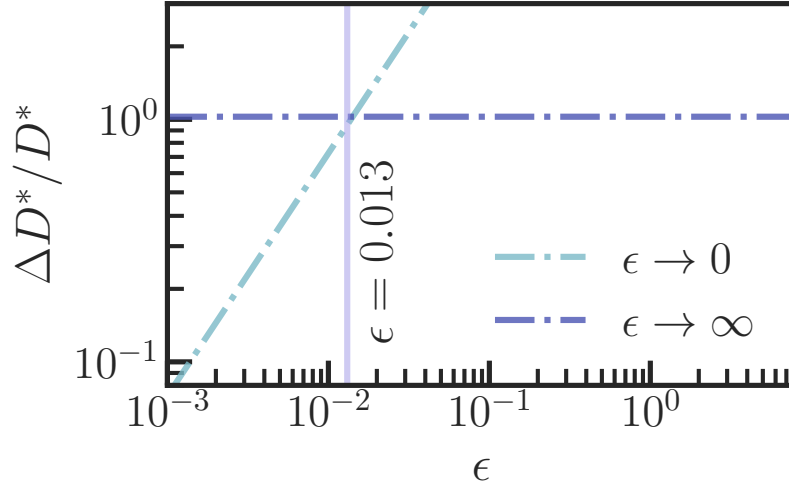


Figure 1.7: Estimation of the relative enhancement of the long-time diffusion coefficient of a Brownian colloid of radius $a = 100\text{nm}$ in a periodic potential of amplitude $\Delta U = 4k_{\text{B}}T$ and spatial period $\lambda = 100\text{nm}$ at a distance $d = 10\text{nm}$ from membrane of elastic modulus $E = 1\text{kPa}$ or equivalent dimensionless compliance $\epsilon = 0.013$. The hydrodynamic constant μ_{12} is estimated using Oseen's model. The two asymptotes of the very rigid $\epsilon \rightarrow 0$ and very soft $\epsilon \rightarrow \infty$ are shown in dash-dotted cyan and blue lines respectively. For this system, the expected value of the relative enhancement of the long-time diffusion coefficient of the Brownian colloid is of about $\frac{\Delta D^*}{D^*} \approx 100\%$.

we have implemented numerical simulations to verify these predictions. We have found that the coupling of the membrane to the Brownian colloid always increases its long-time diffusion constant given the framework of our model.

However, this systematic increase should be heavily modified by the introduction of more physical ingredients in the model. For instance, the attentive reader may have noticed our mobility matrix is not position-dependant, which hydrodynamically speaking would involve colloids quite far apart compared to their respective radii, rendering the coupling extremely weak. Furthermore, the system is so far in only one dimension, which should modify quantitatively the expected enhancement on the diffusion coefficient of the Brownian colloid. We also work in overdamped regime, but acknowledge the interest of keeping mass in the system so as to probe at added mass effects which are currently being investigated within our group.

Nevertheless, the methodology presented in this Chapter as well as the numerical methods are solid and, as the time of writing of this manuscript, being extended to some cases mentioned above.

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